

Most Important Pandas String Functions

Priority	Function	What does it do?	One-line Example
★★★★★	<code>str.lower()</code>	Convert text to lowercase	<code>df["Name"].str.lower()</code>
★★★★★	<code>str.upper()</code>	Convert text to uppercase	<code>df["Name"].str.upper()</code>
★★★★★	<code>str.title()</code>	Capitalize every word	<code>df["Name"].str.title()</code>
★★★★★	<code>str.strip()</code>	Remove spaces from both ends	<code>df["Name"].str.strip()</code>
★★★★★	<code>str.lstrip()</code>	Remove left spaces	<code>df["Name"].str.lstrip()</code>
★★★★★	<code>str.rstrip()</code>	Remove right spaces	<code>df["Name"].str.rstrip()</code>
★★★★★	<code>str.len()</code>	Find string length	<code>df["Name"].str.len()</code>
★★★★★	<code>str.replace()</code>	Replace text	<code>df["City"].str.replace("Bombay","Mumbai")</code>
★★★★★	<code>str.contains()</code>	Search for text	<code>df["Name"].str.contains("Rohit")</code>
★★★★★	<code>str.startswith()</code>	Check starting text	<code>df["Name"].str.startswith("R")</code>
★★★★★	<code>str.endswith()</code>	Check ending text	<code>df["Email"].str.endswith(".com")</code>
★★★★★	<code>str.count()</code>	Count occurrences	<code>df["Text"].str.count("a")</code>
★★★★★	<code>str.find()</code>	Find first occurrence	<code>df["Name"].str.find("Sharma")</code>
★★★★★	<code>str.split()</code>	Split string	<code>df["Name"].str.split(" ")</code>
★★★★★	<code>str.rsplit()</code>	Split from right	<code>df["Path"].str.rsplit("/",1)</code>
★★★★★	<code>str.cat()</code>	Join string columns	<code>df["First"].str.cat(df["Last"], sep=" ")</code>
★★★★★	<code>str.isalpha()</code>	Check only letters	<code>df["Name"].str.isalpha()</code>
★★★★★	<code>str.isdigit()</code>	Check only digits	<code>df["ID"].str.isdigit()</code>
★★★★★	<code>str.isalnum()</code>	Check letters and digits	<code>df["Code"].str.isalnum()</code>
★★★★★	<code>str.islower()</code>	Check lowercase	<code>df["Text"].str.islower()</code>
★★★★★	<code>str.isupper()</code>	Check uppercase	<code>df["Text"].str.isupper()</code>
★★★★★	<code>str.extract()</code>	Extract regex match	<code>df["Email"].str.extract(r"(\w+)")</code>

Priority	Function	What does it do?	One-line Example
★★★★	str.findall()	Find all regex matches	df["Text"].str.findall(r"\w+")
★★★★	str.slice()	Extract substring	df["Name"].str.slice(0,5)
★★★	str.zfill()	Pad with leading zeros	df["ID"].str.zfill(5)
★★★	str.get()	Get character by index	df["Name"].str.get(0)

Function	use
str.capitalize()	title() is more commonly used.
str.swapcase()	Rarely needed in data analysis.
str.repeat()	Rare practical use.
str.pad()	Mostly for formatting reports.
str.center()	Text formatting only.
str.ljust()	Formatting only.
str.rjust()	Formatting only.
str.partition()	split() is used more often.
str.rpartition()	Rarely required.
str.join()	Python's built-in join() is more common.
str.match()	Used only for regex validation.
str.fullmatch()	Used for strict regex validation.
str.extractall()	Needed only when extracting multiple regex matches.
str.isnumeric()	Similar to isdigit() in most practical datasets.
str.isdecimal()	Special Unicode use cases.
str.istitle()	Rare validation requirement.
str.isspace()	Rarely used.
str.encode()	Used in encoding/decoding tasks, not everyday data cleaning.
str.decode()	Mainly for byte data.
str.normalize()	Mostly for multilingual Unicode text.

Function	use
<code>str.wrap()</code>	Report formatting.
<code>str.translate()</code>	Advanced character replacement.
<code>str.rfind()</code>	Less common than <code>find()</code> .
<code>str.slice_replace()</code>	Niche editing operation.